

08/04/2024

Phase-I
CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

CF

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 6

Time : 3 Hrs. 20 Mins.

Answers

1. (4)	41. (2)	81. (1)	121. (3)	161. (3)
2. (3)	42. (2)	82. (3)	122. (4)	162. (1)
3. (2)	43. (2)	83. (3)	123. (2)	163. (1)
4. (4)	44. (4)	84. (1)	124. (4)	164. (4)
5. (2)	45. (2)	85. (4)	125. (3)	165. (4)
6. (1)	46. (4)	86. (4)	126. (4)	166. (2)
7. (4)	47. (1)	87. (4)	127. (4)	167. (3)
8. (2)	48. (2)	88. (2)	128. (2)	168. (4)
9. (3)	49. (2)	89. (3)	129. (1)	169. (4)
10. (1)	50. (3)	90. (2)	130. (1)	170. (2)
11. (1)	51. (1)	91. (4)	131. (3)	171. (4)
12. (3)	52. (3)	92. (1)	132. (1)	172. (4)
13. (4)	53. (2)	93. (2)	133. (3)	173. (4)
14. (3)	54. (1)	94. (4)	134. (2)	174. (4)
15. (1)	55. (3)	95. (4)	135. (2)	175. (3)
16. (2)	56. (4)	96. (1)	136. (2)	176. (3)
17. (4)	57. (4)	97. (4)	137. (4)	177. (4)
18. (3)	58. (4)	98. (1)	138. (3)	178. (2)
19. (3)	59. (1)	99. (1)	139. (1)	179. (3)
20. (3)	60. (4)	100. (2)	140. (3)	180. (2)
21. (4)	61. (3)	101. (3)	141. (1)	181. (3)
22. (4)	62. (3)	102. (2)	142. (3)	182. (3)
23. (3)	63. (4)	103. (4)	143. (1)	183. (3)
24. (3)	64. (2)	104. (3)	144. (3)	184. (1)
25. (2)	65. (4)	105. (2)	145. (1)	185. (3)
26. (4)	66. (1)	106. (4)	146. (2)	186. (2)
27. (2)	67. (2)	107. (2)	147. (3)	187. (3)
28. (4)	68. (4)	108. (3)	148. (3)	188. (1)
29. (1)	69. (3)	109. (4)	149. (2)	189. (2)
30. (2)	70. (1)	110. (1)	150. (4)	190. (1)
31. (2)	71. (1)	111. (2)	151. (4)	191. (4)
32. (4)	72. (3)	112. (4)	152. (4)	192. (4)
33. (3)	73. (1)	113. (2)	153. (1)	193. (4)
34. (2)	74. (3)	114. (3)	154. (2)	194. (2)
35. (4)	75. (3)	115. (3)	155. (3)	195. (1)
36. (1)	76. (4)	116. (1)	156. (3)	196. (3)
37. (3)	77. (3)	117. (4)	157. (2)	197. (2)
38. (2)	78. (4)	118. (1)	158. (3)	198. (1)
39. (3)	79. (4)	119. (3)	159. (2)	199. (4)
40. (3)	80. (3)	120. (1)	160. (3)	200. (1)



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 6

Time : 3 Hrs. 20 Mins.

Answers and Solutions

PHYSICS

SECTION - A

1. Answer (4)

Magnetic moment = $i \times \pi R^2$

So if radius of circular loop is doubled, then its magnetic moment becomes four times.

2. Answer (3)

A moving coil galvanometer uses the principle of magnetic effect of current for its working.

3. Answer (2)

$$B = \frac{\mu_0 i R^2}{2(x^2 + R^2)^{3/2}}$$

At centre of the coil:

$$B_0 = \frac{\mu_0 i}{2R}$$

$$\therefore \frac{\mu_0 i}{2R} \times \frac{1}{27} = \frac{\mu_0 i R^2}{2(x^2 + R^2)^{3/2}}$$

$$(x^2 + R^2)^{3/2} = 27R^3$$

$$x^2 + R^2 = (27R^3)^{2/3}$$

$$x^2 + R^2 = 9R^2 \Rightarrow 8R^2 = x^2 \Rightarrow x = 2\sqrt{2}R$$

4. Answer (4)

The direction of $i\vec{dl} \times \vec{r}$ vector will give the direction of magnetic field. Here the direction of magnetic field is towards south.

5. Answer (2)

Time period in magnetic field is given by $T = \frac{2\pi m}{qB}$,

which is independent of speed of charge.

6. Answer (1)

$$I = \frac{dq}{dt} = \frac{t^3}{3} + 2t^2$$

$$\int_0^q dq = \int_0^3 \frac{1}{3} t^3 dt + \int_0^3 2t^2 dt$$

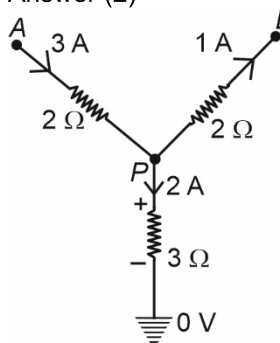
$$q = \frac{1}{3} \times \frac{t^4}{4} \Big|_0^3 + 2 \frac{t^3}{3} \Big|_0^3$$

$$q = \frac{27}{4} + 18 = 24.75 \text{ C}$$

7. Answer (4)

$$I = \frac{V}{R_{eq}} = \frac{6-2}{1+2+3} = \frac{4}{6} = \frac{2}{3} \text{ A}$$

8. Answer (2)



Potential difference across 3Ω resistor
 $= 3 \times 2 = 6 \text{ V}$

$$\therefore V_P = 6 \text{ V}$$

9. Answer (3)

Work done by battery = $\Delta V \times q$

$$W_b = 1.5 \times 6 \text{ mJ}$$

$$= 9 \text{ mJ}$$

10. Answer (1)

- Drift velocity of electrons $\rightarrow V_d$
Thermal velocity of electrons $\rightarrow V_t$
 $V_d \ll V_t$
- Resistance of conductors increase with increase in temperature
- In a cell, current can flow in either direction depending on whether it is charging or discharging.

11. Answer (1)

Unit of intensity of magnetisation is $A\ m^{-1}$

12. Answer (3)

$$\vec{\tau} = \vec{\mu} \times \vec{B} \Rightarrow \tau = \mu B \sin \theta$$

$$\therefore \tau = 20 \times \frac{5}{100} \times 10^{-3} \times \sin 30^\circ$$

$$= 5 \times 10^{-4} \text{ N m}$$

13. Answer (4)

- Whenever there is change in magnetic flux through a coil, there will be induced emf.
- Lenz law is based on the principle of conservation of energy.

14. Answer (3)

Emf or potential difference across an induction coil is given by

$$\Delta V = -\frac{L di}{dt}$$

$$\Delta V = -2 \times 2.5 = -5 \text{ V}$$

15. Answer (1)

$$\phi_2 = M i_1$$

Where M is coefficient of mutual inductance

$$\Delta \phi = M \Delta i \Rightarrow 0.8 = M \times 2$$

$$M = 0.4 \text{ H}$$

16. Answer (2)

Potential difference across R is 4 V, hence

$$\Delta V = iR$$

$$4 = i \times R \quad \dots(i)$$

$$8 = iR + 200 \times i \quad \text{from Kirchhoff's law}$$

$$8 - 4 = 200i \Rightarrow i = \frac{4}{200} \quad \dots(ii)$$

$$\therefore 4 = iR \Rightarrow 4 = \frac{4}{200} \times R$$

$$R = 200 \ \Omega$$

17. Answer (4)

$$\text{Given } \chi = 699, \mu_r = 1 + \chi = 700$$

$$\mu = \mu_r \mu_0 = 700 \times 4\pi \times 10^{-7} = 28\pi \times 10^{-5}$$

18. Answer (3)

$$|\varepsilon| = \frac{d\phi}{dt} = 3t^2 + 2$$

$$\text{At } t = 3 \text{ s,}$$

$$\varepsilon = 3 \times 3^2 + 2 = 29 \text{ Wb}$$

19. Answer (3)

$$I = nev_0 A$$

$$\frac{I}{2} = nev_1 \frac{A}{4}$$

$$v_1 = \frac{2I}{neA}$$

$$v_1 = 2v_0$$

20. Answer (3)

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

$$R = \frac{\rho \frac{l}{A} \times 2\rho \frac{l}{A}}{\rho \frac{l}{A} + 2\rho \frac{l}{A}}$$

$$\rho_{\text{eff}} \frac{l}{2A} = \frac{2\rho l}{3A}$$

$$\rho_{\text{eff}} = \frac{4\rho}{3}$$

21. Answer (4)

$$\mu = \frac{|\vec{v}_s|}{|\vec{E}|}$$

$$\mu = \frac{[LT^{-1}]}{[MLT^{-2}][AT]}$$

$$\mu = [M^{-1}L^0T^2A]$$

22. Answer (4)

$$\hat{i} = \vec{J} \cdot \vec{A}$$

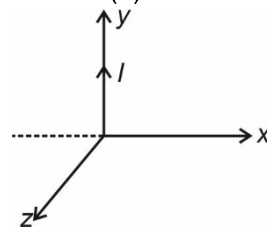
$$= (2 + 12) \times 10^{-6} \text{ A}$$

$$= 14 \ \mu\text{A}$$

23. Answer (3)

Momentum of the particle will change and kinetic energy remains same.

24. Answer (3)



$$\vec{B}_x = \frac{\mu_0 \hat{i}}{2\pi a} (-\hat{j})$$

$$\vec{B}_y = \frac{\mu_0 \hat{I}}{2\pi a}$$

$$\vec{B} = \vec{B}_x + \vec{B}_y$$

25. Answer (2)

$$\vec{F}_m = I(\vec{\ell}_{\text{eff}} \times \vec{B})$$

$$= 2IRB$$

26. Answer (4)

Path of the particle will be helical.

27. Answer (2)

Diamagnetic materials have no permanent magnetic moment

28. Answer (4)

$$B = \frac{\mu_0 M}{4\pi r^3} \sqrt{3 \cos^2 30^\circ + 1}$$

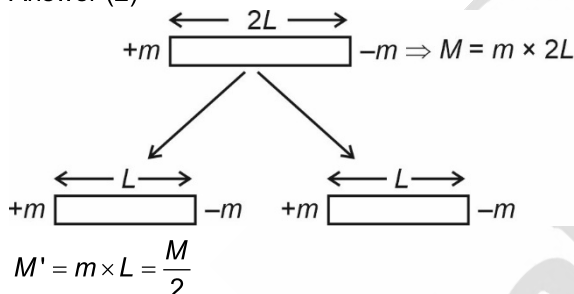
$$= \frac{\sqrt{13}}{8} \frac{\mu_0 M}{\pi r^3}$$

29. Answer (1)

Self-inductance of a solenoid coil is given by

$$L = \mu_0 n^2 A l$$

30. Answer (2)



31. Answer (2)

$$i = \frac{d\phi}{dt \times R}$$

$$\phi = R \int i dt$$

$$\phi = 5 \times 6 \times 0.2 \times \frac{1}{2} = 3$$

32. Answer (4)

$$\varepsilon = \frac{B\omega l^2}{2}$$

$$60 = \frac{B \times 100 \times 4}{2}$$

$$B = \frac{30}{100}$$

$$B = 0.3 \text{ T}$$

33. Answer (3)

Potential difference in this condition will be

$$\varepsilon = Blv = 2 \times \frac{10}{100} \times 3$$

$$\varepsilon = 0.6 \text{ V}$$

34. Answer (2)

Mutual inductance of two coils can be increased by either increasing the number of turns in the coils or winding the coils on steel core or both.

35. Answer (4)

For parallel combination

$$\frac{1}{L'} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \Rightarrow L' = \frac{2}{3} \text{ H}$$

For series combination

$$L_{eq} = 2 \text{ H} + \frac{2}{3} \text{ H} = \frac{8}{3} \text{ H}$$

SECTION-B

36. Answer (1)

$$V_J = \frac{\frac{10}{1} + \frac{20}{2} + \frac{30}{3} + \frac{40}{4}}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}$$

$$V_J = \frac{40}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}$$

$$V_J = \frac{40 \times 12}{12 + 6 + 4 + 3} = \frac{480}{25} \text{ V} = 19.2 \text{ V}$$

37. Answer (3)

$$P_1 = P_2$$

$$i_1^2 \cdot R_1 = i_2^2 R_2$$

$$\frac{\varepsilon^2}{(R_1 + r)^2} R_1 = \frac{\varepsilon^2}{(R_2 + r)^2} R_2$$

$$R_1(R_2 + r)^2 = R_2(R_1 + r)^2$$

$$\sqrt{R_1}(R_2 + r) = \sqrt{R_2}(R_1 + r)$$

$$\sqrt{R_1 R_2}(\sqrt{R_2} - \sqrt{R_1}) = (\sqrt{R_2} - \sqrt{R_1})r$$

$$r = \sqrt{R_1 R_2}$$

38. Answer (2)

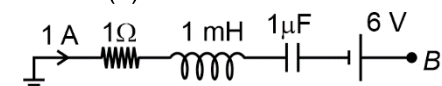
$$r = \frac{P}{qB}$$

$$\text{So, } r_d : r_\alpha = 2 : 1$$

39. Answer (3)

Superconductors are perfect diamagnetic material

40. Answer (3)



$$\text{KVL : } 0 - 1 - 1 + V_C + 6 = 10$$

$$V_C = 6 \text{ V}$$

$$Q = CV_C = 6 \mu\text{C}$$

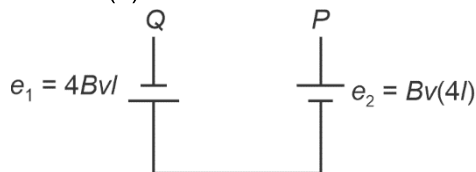
41. Answer (2)

$$F = 2 \times 10^{-7} \times 2 \times 8 \left[\frac{1}{2} - \frac{1}{10} \right]$$

$$= 1.28 \times 10^{-6} \text{ N}$$

$$F_{\text{repulsive}} > F_{\text{attractive}}$$

42. Answer (2)



$$V_{AP} = 8BvI$$

43. Answer (2)

$$P = \frac{2\pi m}{qB} v \cos 37^\circ$$

$$r = \frac{mv \sin 37^\circ}{qB} = \frac{P}{2\pi} \tan 37^\circ$$

$$r = \frac{3P}{8\pi}$$

44. Answer (4)

Magnetic moment of the given loop

$$\vec{M} = Ni\vec{A}$$

$$\vec{M} = 3\pi \left(\frac{5}{100} \right)^2 \times 50 \hat{k}$$

$$= \frac{3\pi}{8} \hat{k}$$

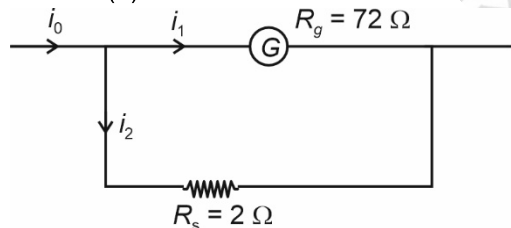
Torque on the loop

$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$= \left(\frac{3\pi}{8} \hat{k} \right) \times 2\hat{i}$$

$$= \frac{3\pi}{4} \hat{j}$$

45. Answer (2)



$$i_1 R_g = i_2 R_s$$

$$i_1 \times 72 = i_2 \times 2$$

$$i_1 = \frac{i_2}{36}$$

$$i_1 + i_2 = i_0$$

$$\frac{i_1}{i_0} = \frac{i_1}{i_1 + 36i_1} = \frac{1}{37}$$

46. Answer (4)

Above curie temperature, ferromagnetic material becomes paramagnetic.

47. Answer (1)

- Throughout the conductor current will remain same.

- Current density = $\frac{\text{Current}}{\text{Area}}$

Since current is same and area is variable, therefore current density is variable.

- $I = neAV_d \Rightarrow V_d \propto \frac{1}{A}$

- $J = \sigma E$

Since J is variable, there E is variable.

48. Answer (2)

$$R = \frac{\rho l^2}{V}, \text{ where } V \text{ is the volume of the wire}$$

$$\frac{\Delta R}{R} = \frac{2\Delta l}{l} \Rightarrow \frac{\Delta R}{R} \times 100 = 2 \times 0.3 = 0.6\%$$

49. Answer (2)

$$\phi = BAN, \quad \varepsilon = \frac{d\phi}{dt}, \quad \varepsilon = iR$$

$$iR = \left(\frac{dB}{dt} \right) AN \Rightarrow i \times 20 = 10^3 \times 20 \times 10^{-4} \times 50$$

$$i = 5 \text{ A}$$

50. Answer (3)

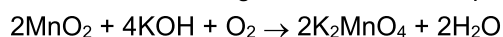
According to lenz law, the emf will be induced such that it will oppose the change in flux.

$$\theta = \frac{230}{6} = 38.3^\circ \text{C}$$

CHEMISTRY

SECTION - A

51. Answer (1)

MnO₂ on reaction with KOH and KNO₃ gives K₂MnO₄ which is a green coloured compound.

52. Answer (3)

German silver contains Cu(25-30%), Zn(25-30%), Ni(40-50%)

53. Answer (2)

Red phosphorus does not glow in dark.

54. Answer (1)

The transition metal ions which contain electron(s) in d-orbitals (except d^{10}) will show colour in aqueous medium.

Zn^{2+} : Colourless

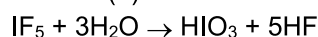
Cr^{3+} : Violet

Ti^{3+} : Purple

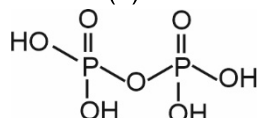
Sc^{3+} : Colourless

Ni^{2+} : Green

55. Answer (3)



56. Answer (4)

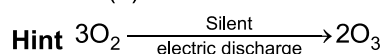


Number of P-OH bonds is four.

57. Answer (4)

For Haber's process, the optimum conditions for the production of ammonia are 200 atm pressure and about 700 K temperature.

58. Answer (4)



Sol.

- Deacon's process $\rightarrow$ Preparation of Cl_2
- Contact process $\rightarrow$ Preparation of H_2SO_4
- Ostwald's process $\rightarrow$ Preparation of HNO_3

59. Answer (1)

- N_2O is neutral oxide
- Negative electron gain enthalpy of selenium is greater than oxygen.

60. Answer (4)

Metal	Enthalpy of atomization (kJ mol^{-1})
Sc	326
Ti	473
Mn	281
V	515

61. Answer (3)

Most common oxidation states of copper and titanium respectively are +2 and +4

62. Answer (3)

In $\text{Ni}(\text{CO})_4$, central atom is sp^3 hybridised and the complex is tetrahedral in shape.

63. Answer (4)

Ions	Ionic radii (pm)
Nd^{3+}	99
Gd^{3+}	94
Er^{3+}	88

Higher is the effective nuclear charge, smaller will be size of lanthanoid ion.

64. Answer (2)

Compounds	Bond angle
H_2O	104°
H_2S	92
H_2Se	91
H_2Te	90

65. Answer (4)

Boiling point of ammonia is greater than phosphine because ammonia molecules are associated with each other by intermolecular hydrogen bonding.

66. Answer (1)

Stronger is the ligand field, higher will be the value of Δ_0 , and higher will be the energy absorbed hence shorter will be the value of wavelength.

67. Answer (2)

$\text{XeF}_4 \rightarrow sp^3d^2$, square planar

$\text{XeOF}_4 \rightarrow sp^3d^2$, square pyramidal

$\text{XeO}_2\text{F}_2 \rightarrow sp^3d$, see-saw

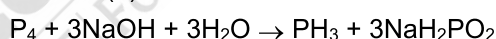
$\text{XeO}_3 \rightarrow sp^3$, pyramidal

68. Answer (4)

$E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ is positive, hence copper can not liberate

H_2 gas on reaction with dilute H_2SO_4 .

69. Answer (3)



70. Answer (1)

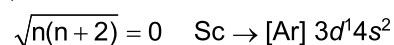
$[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ is inner orbital complex. It is diamagnetic in nature.

71. Answer (1)

Metal cation	$\Delta_{\text{hyd}}H^\circ$ (kJ mol^{-1})
Ni^{2+}	-2121
Ti^{2+}	-1866
Cr^{2+}	-1925
V^{2+}	-1895

72. Answer (3)

$\mu = 0$ $n = \text{number of unpaired electron.}$



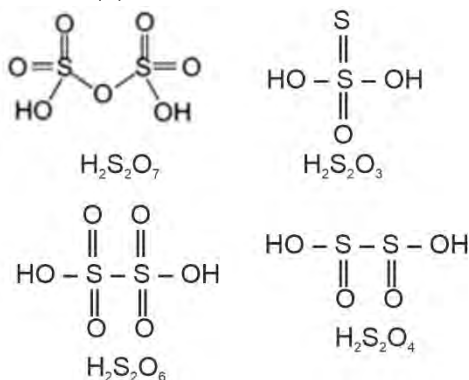
73. Answer (1)

$\Delta_f H^\circ (\text{NH}_3) = -46.1 \text{ kJ mol}^{-1}$, $\Delta_f H^\circ$ of other hydrides are positive.

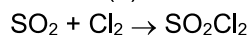
74. Answer (3)



75. Answer (3)



76. Answer (4)



Charcoal is used as catalyst.

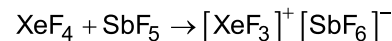
77. Answer (3)

Wilkinson catalyst: $[(\text{Ph}_3\text{P})_3\text{RhCl}]$

78. Answer (4)

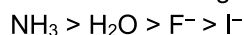
Higher is the charge on metal cation and stronger is the ligand field strength, higher will be the value of crystal field splitting energy, Δ_0 .

79. Answer (4)



80. Answer (3)

Correct order of ligand field strength



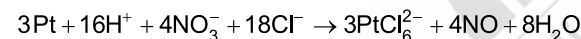
81. Answer (1)

Lesser is the electron density on the metal, lesser is the back donation from metal to ligand and longer will be the metal-carbon bond length.

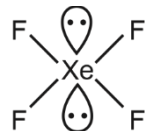
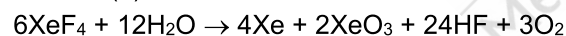
82. Answer (3)



83. Answer (3)



84. Answer (1)



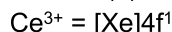
Square planar

85. Answer (4)

Species	Colour
CrO_4^{2-}	Yellow
$\text{Cr}_2\text{O}_7^{2-}$	Orange
MnO_4^-	Purple
MnO_4^{2-}	Green

SECTION - B

86. Answer (4)

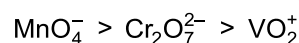


87. Answer (4)

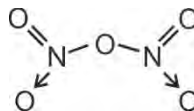
 $\text{K}_3[\text{Fe}(\text{CN})_6]$: Potassium hexacyanidoferrate(III)

88. Answer (2)

Correct order of oxidising power is



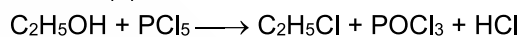
89. Answer (3)



90. Answer (2)

$$\Delta_0 = \frac{9}{4} \Delta_t$$

91. Answer (4)



92. Answer (1)

For $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Primary valency of Co = 3

Secondary valency of Co = 6

93. Answer (2)

Total 3 isomers are possible for the co-ordination complex $[\text{MABCD}]$.

94. Answer (4)

Rhombic sulphur is insoluble in water but dissolves to some extent in benzene.

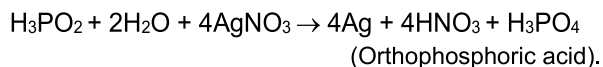
95. Answer (4)

Order of IE_1 : $\text{Zn} > \text{Fe} > \text{Co} > \text{Ni}$

96. Answer (1)

More is the number of ions in the solution, more will be the molar conductance.

97. Answer (4)



98. Answer (1)

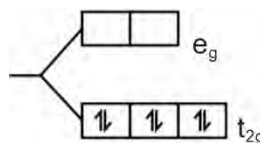
 CrO – basic Cr_2O_3 – Amphoteric V_2O_5 – Amphoteric Mn_2O_7 - Acidic

99. Answer (1)

Complex species	No. of unpaired electron	$\mu = \sqrt{n(n+2)}$ BM
$[\text{Fe}(\text{CN})_6]^{3-}$	1	$\mu = 1.732$ BM
$[\text{NiCl}_4]^{2-}$	2	$\mu = 2.82$ BM
$[\text{Co}(\text{NH}_3)_6]^{3+}$	0	$\mu = 0$ BM
$[\text{FeF}_6]^{3-}$	5	$\mu = 5.92$ BM

100. Answer (2)

CN⁻ is a strong field ligand so, pairing will take place.



BOTANY

SECTION - A

101. Answer (3)

The basis of DNA fingerprinting is VNTR.

102. Answer (2)

In human genome, chromosome 1 has most genes i.e. 2968.

103. Answer (4)

Regulatory gene is a constitutive gene which is always functional.

104. Answer (3)

AUG has dual function, it codes for methionine and it also acts as an initiator codon.

105. Answer (2)

George Gamow, a physicist argued that since there are only 4 bases and if they have to code for 20 amino acids, the genetic code should constitute a combination of bases.

106. Answer (4)

RNA polymerase III catalyses the synthesis of tRNA, ScRNA, 5S rRNA and SnRNA.

RNA polymerase II synthesises hnRNA.

107. Answer (2)

The mRNA provides the template, tRNA brings amino acids and reads the genetic code and rRNA plays structural and catalytic role during translation.

108. Answer (3)

There is a single DNA dependent RNA polymerase that catalyses transcription of all types of RNA in bacteria.

109. Answer (4)

Splicing is the process of removal of introns and joining of exons in a defined order in eukaryotes.

110. Answer (1)

For long DNA molecule, since the two strands of DNA cannot be separated in its entire length (due to very high energy requirement) the replication occur within a small opening of DNA helix, referred to as replication fork.

111. Answer (2)

RNA is nucleic acid which can act as structural molecule.

112. Answer (4)

DNA as an acidic substance present in nucleus was named 'Nuclein' by Friedrich Meischer.

113. Answer (2)

$\phi \times 174$ bacteriophage has only 5386 bases in its genetic material whereas bacteriophage λ , *E. coli* and human genome have 48502, 4.6×10^6 and 3.3×10^9 base pairs respectively in their genetic materials.

114. Answer (3)

Bacterial DNA is double stranded and circular. In circular DNA number of phosphodiester bonds is equal to the number of bases.

115. Answer (3)

Adenine, Guanine and Cytosine are present in both DNA and RNA. Thymine is present in DNA and Uracil is present in RNA.

116. Answer (1)

Two nucleotides are linked through 3' – 5' phosphodiester linkage to form a dinucleotide.

117. Answer (4)

In DNA, the plane of one base pair stacks over the other. This, in addition to H-bonds, confers stability of the helical structure.

118. Answer (1)

In dsDNA, purines and pyrimidines occur in equal amounts.

119. Answer (3)

Genetics is the science of heredity and variation. Inheritance is the process by which characters are passed on from parent to offspring. It is the basis of heredity.

120. Answer (1)

Alleles are slightly different forms of the same gene.

F₁ resembles both the parents in case of co-dominance, is in-between in case of incomplete dominance.

121. Answer (3)

A typical Mendelian dihybrid cross will result in 9 : 3 : 3 : 1 phenotypic ratio in F₂ generation.

122. Answer (4)

Types of gamete = 2^n

Where n = degree of heterozygosity

So, 2^3 or total eight different types of gamete will be produced by the individual.

123. Answer (2)

Ratio of different genotypes to phenotypes regarding ABO blood group in human population is

$$\frac{6}{4} = \frac{3}{2} \text{ or } 3 : 2.$$

124. Answer (4)

Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and mapped their position on chromosome.

125. Answer (3)

Phenylketonuria affected individual lacks an enzyme that converts amino acid phenylalanine into tyrosine. As a result of this phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives.

126. Answer (4)

Sickle cell anaemia is an autosome recessive trait. If parents are carrier or heterozygous, then their offspring can express in the ratio of 1 homozygous for disease, 2 heterozygous carrier and 1 homozygous normal.

127. Answer (4)

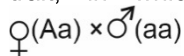
Down's syndrome is caused due to trisomy of chromosome 21. It is an autosomal condition seen in both males and females.

128. Answer (2)

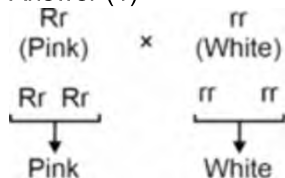
If size of starch grains in pea is considered as phenotype, Bb alleles show incomplete dominance.

129. Answer (1)

The given pedigree shows autosomal dominant trait, in which the parental combination is



130. Answer (1)



No red colored flower will be produced in the above cross.

131. Answer (3)

AB blood group in humans is an example of co-dominance.

When I^A and I^B are present together they both express their own types of sugar. This is because of co-dominance.

Flower colour in dog flower and starch grain size in pea seeds are examples of incomplete dominance.

132. Answer (1)

The diagram is called Punnett square and it was developed by a British geneticist, Reginald C. Punnett.

It helps in understanding the production of gametes by the parents, the formation of zygotes and the F_1 and F_2 plants from the diagram.

133. Answer (3)

A true breeding line is one that, having undergone continuous self-pollination, shows the stable trait inheritance and expression for several generations.

Mendel selected 14 true-breeding pea plant varieties as pairs which were similar except for one character with contrasting traits.

134. Answer (2)

Alleles occupy the same loci, as the position, i.e., locus of each gene is fixed on the chromosome.

The two alleles of a gene pair are located on homologous sites of homologous chromosomes.

135. Answer (2)

Chances of a male or a female child is equal, i.e., 1 : 1 in each pregnancy.

SECTION - B

136. Answer (2)

Dominant traits of pea plant are green pod colour, yellow seed colour and violet flower colour.

137. Answer (4)

Plant height in pea plant is a qualitative trait. Kernel colour in wheat, height in human and skin colour in human are quantitative traits.

138. Answer (3)

Monohybrid genotype is heterozygous for genes controlling one character. Therefore, RRPpQQ is genotype showing monohybrid condition.

139. Answer (1)

Morgan dihybrid cross in *Drosophila* lead to the observation that the proportion of parental gene combination were much higher than the non-parental type.

140. Answer (3)

Sickle cell anaemia is caused by the substitution of glutamic acid by valine at the sixth position of the beta globin chain of haemoglobin molecule.

141. Answer (1)

Grasshopper is an example of XO type of sex-determination.

In *Drosophila* XY type of sex determination is seen.

In case of honeybee, sex is based on the number of sets of chromosomes an individual receives.

Sex chromosomes in birds are designated as Z and W, and the male is the homomorphic.

142. Answer (3)

Polygenic traits are not so distinct in their occurrence and are spread across a gradient.

For example, in humans we do not just have tall or short people as two distinct alternatives but a whole range of possible heights.

Such traits are generally controlled by three or more genes and are thus called as polygenic traits.

143. Answer (1)

In haplo-diploid system as seen in honeybee, males (drone) develop from unfertilised egg by means of parthenogenesis.

They are haploid with 16 chromosomes, female (queen or worker) are diploid, with 32 chromosomes.

Males do not have father and thus cannot have sons, but have a grandfather and can have grandsons.

144. Answer (3)

Central dogma in molecular biology states that the genetic information flows from DNA → RNA → protein.

145. Answer (1)

R strain bacteria had been transformed by heat-killed S strain bacteria and became pathogenic.

146. Answer (2)

A molecule that acts as genetic material should provide the scope for slow mutation. It is required for evolution.

147. Answer (3)

Ribozymes are catalytic RNA molecules that catalyze some specific biochemical reactions.

148. Answer (3)

In the third generation, out of eight cells only one cell will have hybrid DNA. Therefore, percent of cell

$$\text{having hybrid DNA} = \frac{100}{8} \% = 12.5\%$$

149. Answer (2)

The discontinuously synthesised fragments of DNA are joined by the enzyme DNA ligase.

150. Answer (4)

Promotor is a DNA sequence that provides binding site for RNA polymerase.

ZOOLOGY

SECTION - A

151. Answer (4)

In Miller's experiment, CH₄, NH₃, H₂ were used in 2 : 1 : 2 ratio along with water vapour at 800°C.

152. Answer (4)

Darwin conjectured that many varieties of finches evolved from the original seed-eating ancestral finches.

153. Answer (1)

Some of the land-reptiles went back into water to evolve into fish-like reptiles probably 200 mya (e.g. *Ichthyosaurs*).

154. Answer (2)

Cervical caps and diaphragms are barriers that are inserted into the female reproductive tract. Multiload 375 is a copper releasing IUD, used by females.

'Nirodh' is a popular brand of condom for the males.

155. Answer (3)

The biggest dinosaur '*Tyrannosaurus rex*' was about 20 feet in height and had huge fearsome dagger-like teeth.

156. Answer (3)

Tubectomy is a surgical method of contraception for females.

157. Answer (2)

Frequency of individuals with dominant allele = p^2 .

$$p^2 \text{ is given as } 0.64 \therefore p = \sqrt{0.64} = 0.8$$

As per Hardy-Weinberg Principle,

$$p + q = 1$$

$$0.8 + q = 1$$

$$q = 1 - 0.8 \Rightarrow 0.2$$

Frequency of heterozygous individuals:

$$2pq = 2 \times 0.8 \times 0.2 \Rightarrow 0.32$$

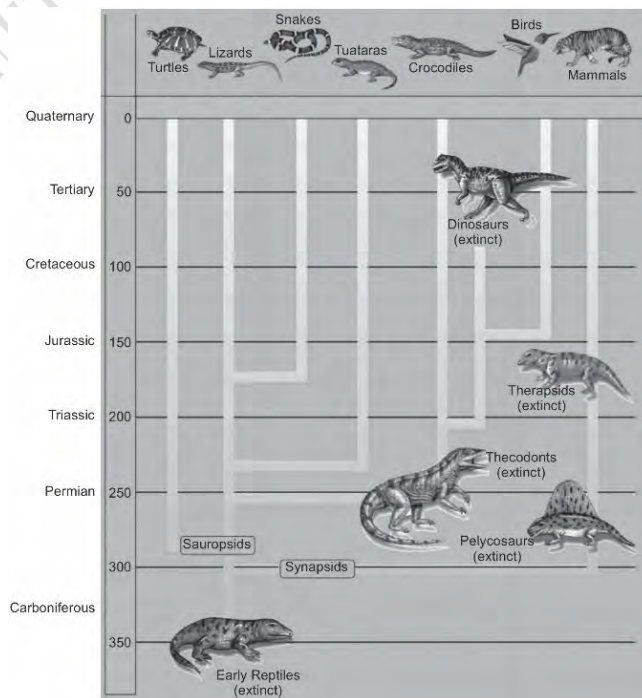
Hence, 32% of people are heterozygous for the same character.

158. Answer (3)

When the peak gets narrower and higher, stabilising type of natural selection occurs. When peak shifts in one direction, that will represent directional selection and when two peaks are formed, disruptive type of natural selection occurs.

159. Answer (2)

Tuatara is a living fossil.



160. Answer (3)
Fossils discovered in Java in 1891, revealed the next stage after *Homo habilis*, i.e., *Homo erectus* about 1.5 mya. *Homo erectus* had a large brain of around 900cc.
161. Answer (3)
Anteater (Placental mammal) and Numbat (Australian marsupial) show convergent evolution.
162. Answer (1)
ICSI - Intra Cytoplasmic Sperm Injection
GIFT - Gamete Intra Fallopian Transfer
IUT - Intra Uterine Transfer
163. Answer (1)
Stochastic process is random and non-directional. It is based on many probabilities but cannot be predicted precisely. Mutations are random and non-directional. Chance events may be anthropogenic and natural.
164. Answer (4)
Vasectomy prevents the transport of sperms and thereby prevents conception.
165. Answer (4)
Dryopithecus was more ape-like and *Ramapithecus* was more man-like.
The Neanderthal man with a brain size of 1400 cc lived in near East and central Asia between 1,00,000 – 40, 000 years back. They used hides to protect their body and buried their dead.
166. Answer (2)
Diaphragms, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus. They are reusable and do not protect the user from contracting STIs such as AIDS.
167. Answer (3)
Infertility cases either due to inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates, could be corrected by artificial insemination (AI) technique.
168. Answer (4)
Embryos with more than 8 blastomeres are transferred into the uterus (IUT-Intra Uterine Transfer).
169. Answer (4)
MTP is essential in certain cases where continuation of pregnancy could be harmful or even fatal either to the mother or to the foetus or both.
170. Answer (2)
Conventional literature tells us about the theory of special creation.
171. Answer (4)
Bryophytes were originated from chlorophyte ancestors. Lycopods were originated from tracheophyte ancestors. Psilophyton were originated from *Rhynia*-type plants.
172. Answer (4)
Vertebrate hearts or brains show homology and divergent evolution.
173. Answer (4)
Pteranodon was a flying reptile.
174. Answer (4)
Due to smoke, tree trunks became dark and predators spotted the white-winged moths against a contrasting background.
175. Answer (3)
Origin of universe occurred around 20 bya. Huge clusters of galaxies comprise the universe. The Big Bang theory attempts to explain to us the origin of universe.
176. Answer (3)
- | | |
|---------------------------|-------------------------|
| <i>Dryopithecus</i> | 15 mya |
| <i>Australopithecines</i> | 2 mya |
| <i>Homo erectus</i> | 1.5 mya |
| <i>Homo sapiens</i> | 75,000-10,000 years ago |
177. Answer (4)
By the time of 500 mya, invertebrates were formed and active.
178. Answer (2)
In 1938, a fish caught in South Africa happened to be a *Coelacanth* which was thought to be extinct.
179. Answer (3)
If the variation in gene frequencies within a population occurs by chance, it is called genetic drift. There would be a gene flow if the gene migration happens multiple times.
180. Answer (2)
Darwin's finches are a good example of adaptive radiation.
The phenomenon of evolution of different species in a given geographical area starting from a point and spreading to other habitats is called adaptive radiation.
181. Answer (3)
Early Greek thinkers thought units of life called spores were transferred to different planets including Earth. Theory of panspermia is related to origin of life.
182. Answer (3)
Ramapithecus was more man-like.
Two mya, *Australopithecines* probably lived in East African grasslands.
Homo erectus probably ate meat.
183. Answer (3)
Except for hepatitis-B, genital herpes and HIV infections, other sexually transmitted diseases are completely curable, if detected early and treated properly.

184. Answer (1)
Progestasert and LNG-20 are hormone releasing IUDs.
Multiload 375 is a copper releasing IUD.

185. Answer (3)
PID, still births, abortions, ectopic pregnancies, infertility or even cancer of the reproductive tract are the complications of STIs.

SECTION - B

186. Answer (2)
Correct sequence of events in the origin of life
Synthesis of organic monomers
↓
Synthesis of organic polymers
↓
Formation of protobionts
↓
Formation of DNA-based genetic systems.
187. Answer (3)
Infections or diseases which are transmitted through sexual intercourse are collectively called sexually transmitted infections (STIs) or venereal diseases (VDs) or reproductive tract infections (RTIs). Diphtheria is a bacterial disease.
188. Answer (1)
The first non-cellular forms of life could have originated 3 billion years back. They would have been giant molecules (RNA, Protein, Polysaccharides, etc.)
189. Answer (2)
The study of history of life forms on Earth is called evolutionary biology.
190. Answer (1)
Platypus is a terrestrial mammal. Whales, dolphins, seals and sea cows are some mammals which live wholly in water.
191. Answer (4)
Similarities in proteins and genes performing a given function among diverse organisms gives clues to common ancestry. Divergent evolution leads to homology.
192. Answer (4)

Human settlement started	–	10,000 years back
--------------------------	---	-------------------

Origin of Earth	–	4.5 billion years ago
Origin of Universe	–	20 billion years ago
Existence of sea weeds and few plants	–	320 mya

193. Answer (4)
In a 32 days menstrual cycle, ovulation occurs on 18th day i.e. = 32-14.
Maximum chances of conception occurs during 14th-21st day.
194. Answer (2)
Only progesterone or in combination with estrogen can inhibit ovulation as well as alter the quality of cervical mucus.
LNG-20, progestasert, OCPs and implants contain progestogens and prevent implantation as well as make cervix hostile to sperms.
195. Answer (1)
In India, there is statutory ban on amniocentesis for sex determination to legally check increasing menace of female foeticide.
196. Answer (3)
According to MTP (Amendment) Act, 2017 a pregnancy may be terminated on certain grounds, within the first 12 weeks of pregnancy on the opinion of one registered medical practitioner. If pregnancy has lasted for more than 12 weeks, but fewer than 24 weeks, two registered medical practitioners, must be of the opinion.
197. Answer (2)
Saheli is a non-steroidal preparation and an oral contraceptive. It is a 'once a week' pill and blocks estrogen receptors in the uterus, preventing egg from getting implanted.
198. Answer (1)
Cu ions releasing from CuT suppress the motility of sperms and their ability to fertilize the ovum.
199. Answer (4)
Increase in number of people in reproductive age is one of the most probable reason for rapid increase in population growth.
200. Answer (1)
In-vitro fertilisation (IVF) is fertilisation outside the female body in almost similar conditions as that in the body followed by Embryo Transfer (ET).